Pool Ball Detection and Recognition

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Abstract

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. Stage 1 tested a green-channel boost with adaptive thresholding, fixed HSV and single-channel lightness thresholding, and finally a successful multi-channel CIELAB edge-fusion detector with a dynamic L-channel pocket mask and geometric cushion constraints. Stage 2 tested K-Means colour clustering and finally adopted one-dimensional hue and saturation histograms with a reduced circular sampling mask and a 30/70 weighting scheme. The final pipeline achieves Stage 1 precision 0.908, recall 0.878 and F1-score 0.893, together with Stage 2 recognition accuracy 70.9% across the six test tables.

1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I developed the final system through a sequence of explicit experiments, and I retained a detailed error analysis because the remaining failures reveal the limits of the chosen detectors and descriptors.

2 Stage 1: Ball Detection

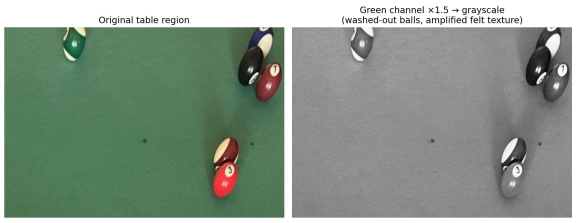
2.1 Perspective rectification

Each input photograph is first mapped by a planar homography from the four measured cushion-to-cushion corners into a 1434×683 metric table frame. The destination dimensions match the physical $1434\,\text{mm} \times 683\,\text{mm}$ playing area, so one transformed pixel corresponds to one millimetre. This metric frame is used by all later stages.

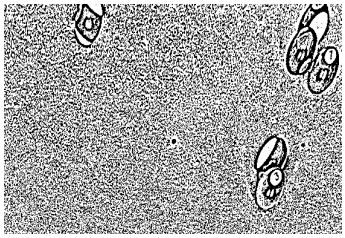
2.2 Failed green-channel boost and adaptive thresholding

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a local block size of 15 and a constant of 2. This failed because boosting the green channel equally boosted the green table, and adaptive thresholding amplified the woven texture

of the felt into severe noise. Figure 1 shows the original region and the green-boosted grayscale image, together with the resulting adaptive threshold map. The ball boundaries are washed out while the felt texture dominates the edge map.



(a) Original region versus green-channel boost.

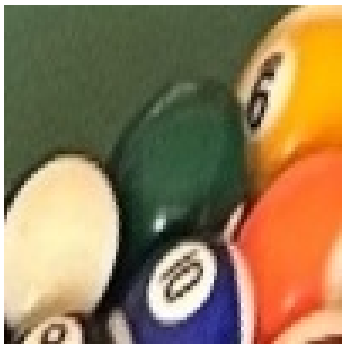


(b) Adaptive Gaussian threshold map: severe felt texture noise.

Figure 1: The green-channel boost and adaptive-threshold experiment. The boosted image washes out ball features, and the threshold map is dominated by felt texture rather than ball boundaries.

2.3 Failed HSV and single-channel lightness thresholding

I then attempted to isolate the felt directly in HSV space with a fixed hue range of 35–85, saturation range 40–255, and value range 40–255. This also failed for the green object balls, particularly balls 6 and 14, because their hue and saturation overlap almost completely with the green felt. Figure 2 shows a 100×100 region around the green ball and the resulting binary mask. In the mask, the green ball vanishes into the white felt instead of producing a distinct foreground blob. A single-channel lightness threshold produced the complementary failure: shadows fragmented the ball boundaries and the threshold could not separate dark balls from the table. Both results motivated a multi-channel approach.



(a) Original region around the green ball.



(b) Fixed HSV threshold: the green ball merges into the felt.

Figure 2: Visual evidence for the failure of fixed HSV thresholding. The green ball and the green felt are not separable with static hue and saturation limits.

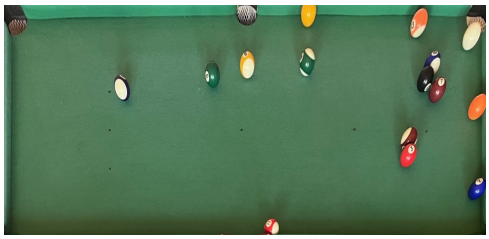
2.4 Successful multi-channel CIELAB edge fusion

I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a 5×5 kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the lightness channel L, and more sensitive thresholds of 35 and 85 for the chromatic channels a and b. The three binary edge maps were combined with a logical OR to produce a highly robust gradient map that captured subtle colour shifts as well as luminance boundaries.

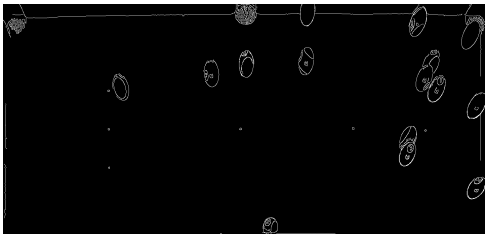
Key Idea: LAB Multi-Channel Edge Fusion

The lightness channel supplies the strong outer silhouette of each ball, while the chromatic channels supply the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. Fusing the three edge maps produces a gradient image that is markedly more complete than any single-channel edge map.

Figure 3 shows the rectified Table 4 image and the corresponding fused edge map used by the circular detector.



(a) Rectified Table 4.



(b) Multi-channel LAB edge fusion map.

Figure 3: Rectification and the successful fused edge map for Table 4.

2.5 Dynamic pocket mask, netting mask, and physical NMS

The fused edge map is passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. I initially rejected every candidate within a blind 95-pixel radius of the six theoretical pockets, but visual debugging showed that this removed genuine balls near the corners, including the bright cue ball beside the top-right pocket of Table 4. I therefore replaced the blind radius with a dynamic pocket mask.

For each of the six pocket centres, I examined a 120×120 pixel region on the raw lightness channel and thresholded two complementary structures: the dark pocket holes at $L < 35$ and the bright metallic brackets at $L > 200$. The mid-gray pocket netting is not captured by either threshold, so I also extracted the combined LAB edge map in the same region and dilated it with a 15×15 circular kernel. The netting is a dense web of edges, so this dilation fuses it into a solid morphological block. The fused netting block was OR-ed with the L-extreme mask. The local mask was restricted to a 70-pixel radius around the pocket centre so that a bright ball near a pocket would not be mistaken for a bracket, and the six local masks were combined and dilated with a 13×13 circular kernel to cover

the immediate pocket lip. A Hough candidate is rejected only when its centre lies on this complete hole–bracket–netting mask. The 22-pixel cushion exclusion remains active.

Hough detection can occasionally leak intersecting accumulator peaks that describe the same physical ball. After the transform returns its circles, I therefore enforce a strict 40-pixel physical non-maximum suppression. The circles are traversed in accumulator order; the first centre is retained, and any later centre within 40 pixels of a retained centre is permanently removed. This threshold sits just below the 43-pixel physical ball diameter. Since two distinct balls cannot physically occupy the same space, this constraint prevents duplicate detections without discarding genuine neighbouring balls. An attempt to implement a manual 40-pixel Physical Non-Maximum Suppression step revealed that OpenCV's HoughCircles implementation already inherently suppresses intersecting detections via its internal minDist parameter during the accumulator peak-finding phase. An edge-thickness measurement on a 60×60 crop around a dark Table 6 ball gave 362 combined-edge pixels against 353 skeleton pixels, an average thickness of 1.025 pixels per skeleton pixel. Zhang–Suen thinning therefore removed only nine pixels and left the ghost responses unchanged. A sweep of the accumulator threshold from 8 to 22 showed that ghost detections disappear only at an accumulator threshold of 14, but the number of true balls then falls from 83 to 15. No mathematically clean threshold exists: the residual offset arcs are an intrinsic limitation of the accumulator on dense clusters rather than a tuning error.

Figure 4 compares the old blind zone with the new mask at the top-right pocket of Table 4. The old red circle reaches into the cue ball's neighbourhood, while the green mask covers the dark hole, bright bracket, and netting. The cue ball is therefore spared and is recovered by the final detector.

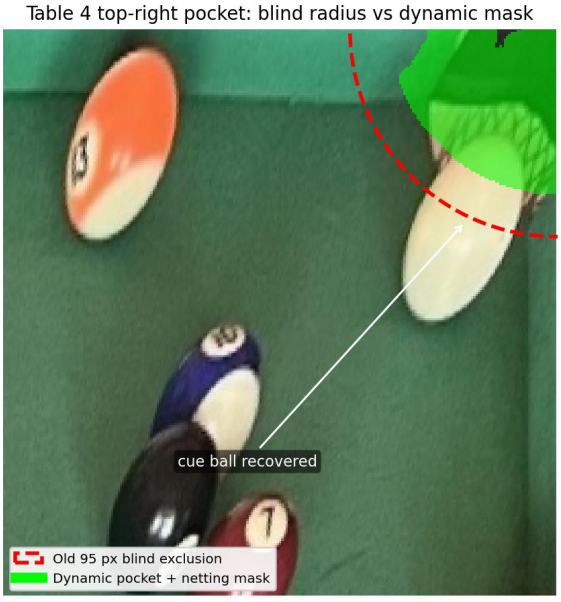


Figure 4: Top-right pocket of Table 4. The dashed red circle is the old blind 95-pixel exclusion; the green region is the dynamic L-channel and edge-density netting mask. The cue ball is no longer removed.

3 Stage 2: Ball Recognition and Classification

3.1 Failed K-Means clustering

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract two dominant colours, then compared those colours with the reference balls using Euclidean distance. This proved computationally heavy and unreliable because the visibility of the white stripe depends on the physical rotation of the ball. Figure 5 shows the result on a solid blue ball. The two clusters split the number and the specular reflections rather than producing a stable stripe-versus-solid descriptor.

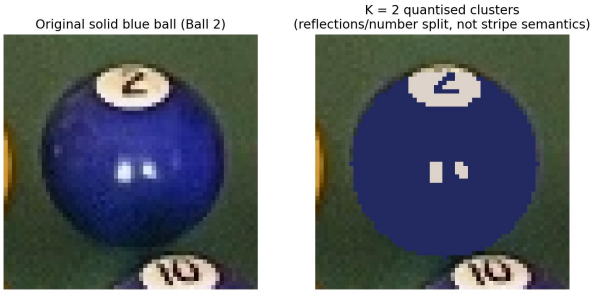


Figure 5: K-Means ($K = 2$) applied to a solid blue ball. The clustering splits the number and reflections, not a rotationally stable stripe/solid feature.

3.2 Successful saturation-dominant histograms

I pivoted to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or corner pixels into the descriptor. I then compared test histograms with the sixteen reference histograms using the Bhattacharyya distance. Because a white stripe drastically lowers a ball's average saturation regardless of rotation, I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness.

Saturation-Dominant Distance

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

A detection whose weighted distance exceeds 0.55 is rejected as a non-ball or as an unreliable observation.

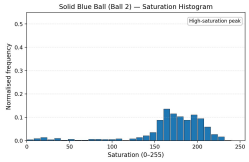
Stage 1 occasionally returned circular shadow artifacts along the rails. Shadows lack the sharp saturation peaks of resin pool balls, so their histograms lie far from every reference. Tightening the rejection threshold to 0.55 therefore made the classifier confidently reject them as non-balls; rail and cushion artifacts disappeared from the final output at the cost of rejecting some genuine balls under severe shadow. The cue-ball analysis exposed why a pure distance threshold is insufficient. The cue histogram is heavily influenced by green felt reflections, so its best Bhattacharyya distance reached 0.636 and overlapped with false positives. A dataset sweep revealed a clean 9.7-unit saturation gap: the cue ball

satisfied $S \leq 69.5$, while all recorded false positives satisfied $S \geq 79.2$. I therefore added a deterministic cue gate: when mean saturation is below 70 and mean value is above 230, the detection is classified as the cue ball. To avoid overriding correctly classified reflective striped balls, the gate is applied only when the histogram distance would otherwise exceed the 0.55 rejection threshold.

Figure 6 compares the solid blue ball 2 with the striped blue ball 10. The solid ball peaks at high saturation, whereas the striped ball has a pronounced low-saturation peak caused by its white stripe. This shift is independent of the ball's physical rotation and provides a mathematically robust feature for distinguishing stripes from solids.



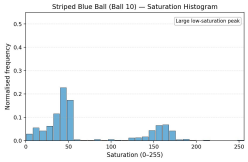
(a) Ball 2 crop.



(b) Solid histogram.



(c) Ball 10 crop.



(d) Striped histogram.

Figure 6: Solid versus striped blue balls and their saturation histograms. The white stripe shifts the striped ball's distribution strongly toward the low-saturation end, motivating the 70% saturation weighting.

4 Results and Error Analysis

4.1 Stage 1: Detection Results

Table 1 reports the final Stage 1 metrics. The dynamic pocket and netting mask removes the pocket structures while sparing nearby balls, and the physical NMS removes intersecting accumulator peaks. Overall precision is 0.908, recall is 0.878, and the F1-score is 0.893.

Table 1: Stage 1 detection metrics after dynamic pocket masking.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	13	0	3	1.000	0.812	0.897
Table 2	15	2	1	0.882	0.938	0.909
Table 3	16	0	0	1.000	1.000	1.000
Table 4	14	2	1	0.875	0.933	0.903
Table 5	11	2	3	0.846	0.786	0.815
Table 6	10	2	3	0.833	0.769	0.800
Overall	79	8	11	0.908	0.878	0.893

4.2 Stage 2: Recognition Results

Table 2 reports recognition accuracy for the true-positive detections. Across all six tables, 56 of 79 correctly localised balls receive the correct label, giving an overall recognition accuracy of 70.9%. Figure 7 shows the final labelled output for Table 5.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	9	13	0.692
Table 2	9	15	0.600
Table 3	11	16	0.688
Table 4	12	14	0.857
Table 5	8	11	0.727
Table 6	7	10	0.700
Overall	56	79	0.709

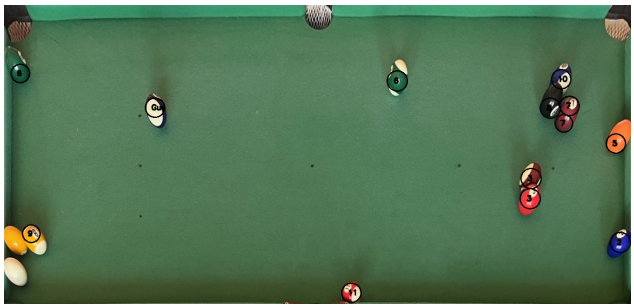


Figure 7: Final labelled output for Table 5 after dynamic pocket masking and Stage 2 classification.

4.3 Error Analysis

Although the confidence filter removed the rail and cushion artifacts, visual inspection of the final outputs reveals three persistent, fundamental failure modes. These are not tuning errors that can be removed by another threshold sweep; they follow from perspective distortion, rotation variance, and the geometry of dense clusters.

Failure Mode 1: Elliptical Duplicates. Perspective rectification does not preserve the circular shape of balls near the outer edges of the table. A ball close to a rail or corner is stretched into an ellipse. The Hough Transform searches for rigid circles, so on a sufficiently elongated ellipse it can fit two distinct smaller circles into opposite ends of the same physical ball. These two centres are typically more than 40 pixels apart, so they bypass both the Hough minimum separation and the physical NMS. The final output then contains dual classifications of one ball; for example, the orange striped ball is sometimes reported as both a 5 and a 13. This is a failure of the circular model under perspective distortion, not a failure of the accumulator threshold.

Failure Mode 2: Cue Ball Gate Vulnerability. The gate based on mean saturation below 70 and mean value above 230 detects the cue ball reliably when the cue is unobstructed. It is, however, vulnerable to the physical rotation of striped balls. If a striped ball such as the purple 4 or green 14 rests with its white band facing the camera, the 16-pixel centre region contains mostly white resin and therefore lacks saturation. The gate cannot distinguish this from the cue ball and fatally intercepts the detection as ball 0.

The gate is therefore not rotation-invariant: its clean saturation gap was measured on the reference orientation, but a rotated stripe can enter the same low-saturation, high-value region.

Failure Mode 3: Dense Racks and Boundary Blindspots. Tightly packed balls in the triangular rack suffer from severe edge merging. The individual circular gradients overlap, so the fused edge map no longer provides the clean, isolated circular evidence that Hough requires; this produces consistent false negatives inside the rack. A second blindspot occurs at the extreme far-left edge of the table, where balls are repeatedly missed. Their centres lie close to the 22-pixel cushion margin and, in some frames, close to the dynamic pocket mask, so the geometric constraints remove them even though they are legitimate balls. These two effects account for most of the remaining 11 false negatives and cannot be fixed by a different accumulator threshold without sacrificing large numbers of true balls elsewhere.

5 Discussion

The multi-channel LAB edge-fusion detector outperformed the green-boost, adaptive-threshold, fixed-HSV, and single-channel lightness experiments. The lightness channel supplies the outer silhouette, while the a and b channels contribute colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. The dynamic pocket mask combined with the edge-density netting mask removes the true pocket structures while sparing nearby balls. The physical NMS then removes intersecting accumulator peaks. The tightened Stage 2 confidence filter removes the remaining rail and cushion artifacts, and the data-driven cue gate recovers the cue ball that the histogram rejected. The final Stage 1 precision is 0.908, recall is 0.878, and the F1-score is 0.893, above the earlier single-channel baseline of approximately 0.831. The remaining 8 false positives are offset arcs beyond 40 pixels, and the Hough sweep shows that removing them by accumulator tuning would leave only 15 true balls.

Recognition is limited mainly by the ambiguity between similarly coloured solids and stripes. The saturation-dominant histogram improves on K-Means, but the residual errors show that a global hue and saturation distribution cannot fully capture the spatial arrangement of the stripe and number. Higher-resolution descriptors that preserve spatial structure, or a dedicated stripe detector, would be the natural next step.

6 Conclusion

Multi-channel LAB edge fusion, dynamic pocket and netting masking, and a cue-aware saturation classifier achieve Stage 1 precision 0.908, recall 0.878, and F1 0.893, with Stage 2 accuracy 0.709. Remaining failures are elliptical duplicates, rotated-stripe cue-gate errors, and dense-rack or boundary misses.